

PROSTHETIC SMART SOCK

Progress Update & Paper Planning

Abdul Qureshi · Calgary Sensor Lab | Project90 · July 2026

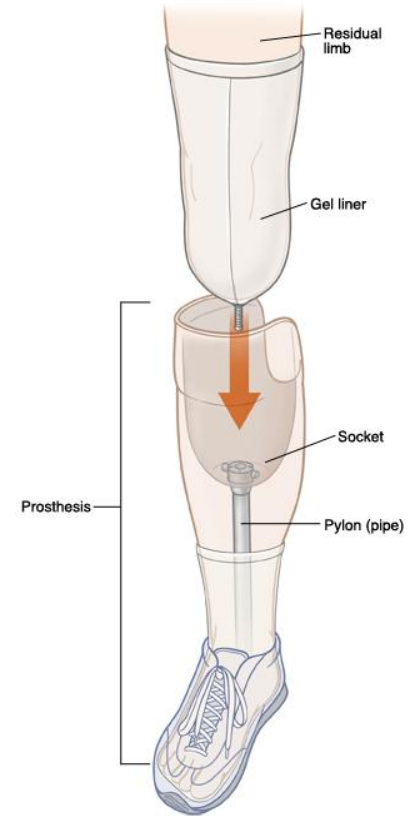


THE PROBLEM

- Designing a comfortable limb prosthetic is a long, painful, & costly journey
- Wrong fit → discomfort, skin breakdown, & pain
- Clinicians rely heavily on subjective patient feedback which provides qualitative evidence rather than direct pressure measurements

OUR AIM

Monitor pressure distribution over the stump for better, more repeatable fitting — not replace clinical judgment.



WHAT OTHERS HAVE DONE

Paper 1: Evaluation of Force Sensing Resistors for the Measurement of Interface Pressures in Lower Limb Prosthetics

FSRs are cheap and thin enough for socket use, but bending, shear, and loading conditions cause large errors — calibrate in the same setup you'll use in the sock.

Paper 2: Preliminary development and validation of a textile-based pressure-sensing system for lower-limb prosthetic sockets

Prosthetic sock sensor with high accuracy. Lab-tested but with no GUI or compact form factor for clinical use.

Paper 3: Textile-based Pressure Sensors for Monitoring Prosthetic-Socket Interfaces

capacitive textile sensors with wireless real-time MATLAB visualization. Good software but hardware needs improvement & no clinical validation completed.

SEARCH THEMES TO FILL IN

- Residual limb socket pressure measurement
- FSR / textile pressure sensor arrays
- Real-time pressure mapping for sockets
- Wearable IoT for rehabilitation / orthotics

Previous studies address parts of this problem, including sensor accuracy, textile integration, and wireless visualization. Our prototype explores how these elements can work together in a low-cost research system.

PROJECT OVERVIEW

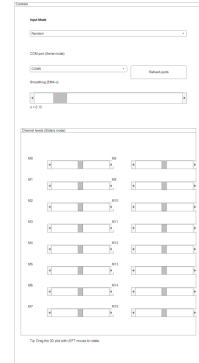
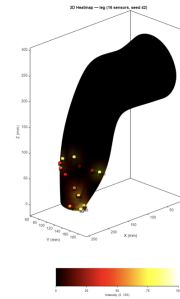
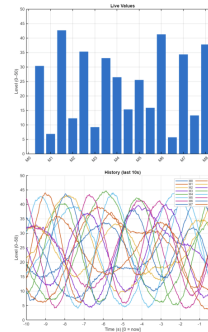


FSR-400 SENSOR ARRAY

MULTIPLEXER PCB

ESP32 + BATTERY PCB

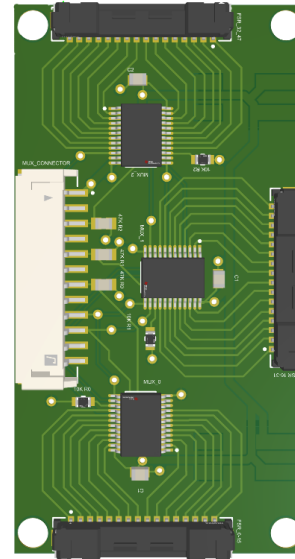
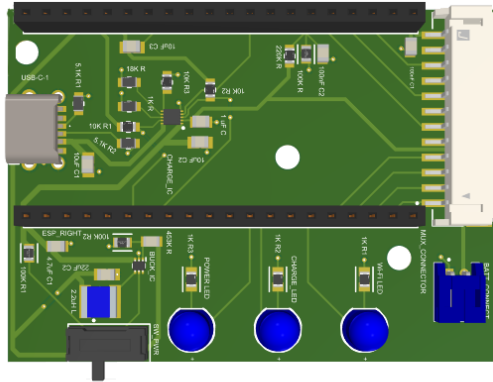
MATLAB GUI



3D Model + PCBs + Wireless Visualization

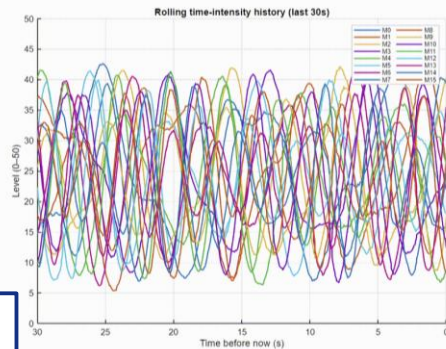
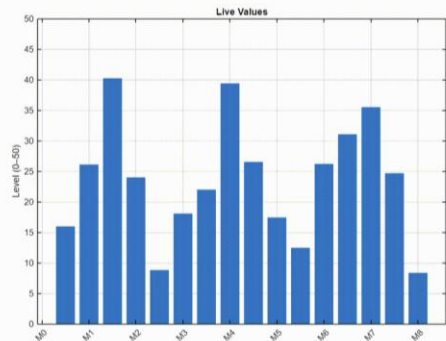
TWO CUSTOM PCBS

Board	Role
Mux PCB	CD74HC4067 16:1 analog multiplexer x3
ESP32 + Battery PCB	Data Acquisition, wireless communication, power distribution

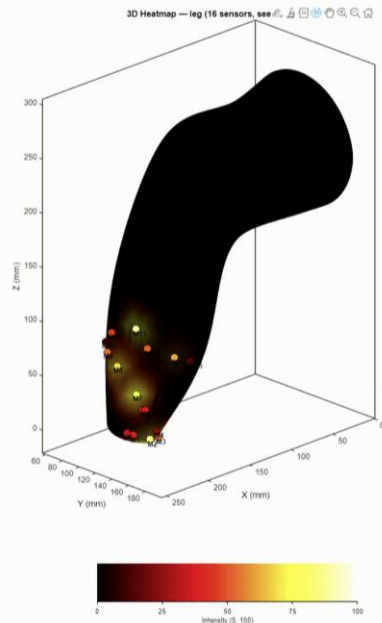


3D REAL-TIME VISUALIZATION

Real-Time
Values

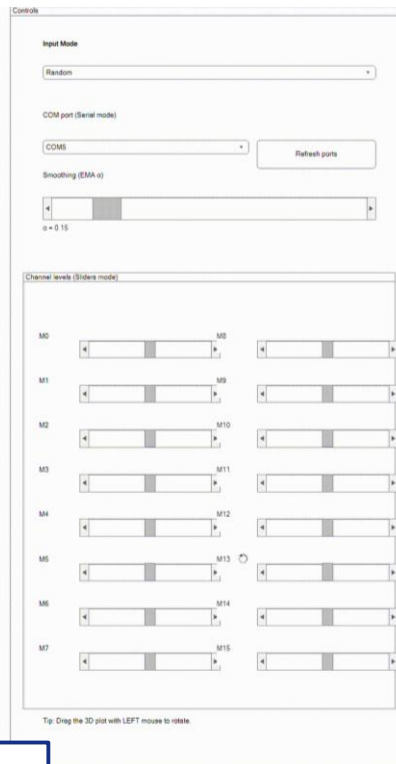


Historical
Values



3D Heatmap

Program
Controls



WORKING VS. NOT VALIDATED

WHAT WORKS TODAY

- ✓ Dual-PCB hardware reads all 16 channels
- ✓ Wi-Fi + serial stream live to MATLAB
- ✓ 3D model: heatmap, bars, history
- ✓ Interactive sensor-site configuration
- ✓ Software-only demo mode (no hardware)

NOT VALIDATED YET

- Pressure-Voltage Calibration
- Clinical trials
- Durability

BENCH TESTS & QUESTIONS FOR THE LAB

Test	Method (one-liner)
Static load cal.	Calibrated masses -> ADC mean $\pm$ SD
Linearity/hysteresis	Ramp load up/down, 5-10 cycles
Mux crosstalk	Press one; read off-channel response
Repeatability	Same load, 10 trials -> CV%
Rate & latency	Measure scan and round-trip latency
Spatial consistency	Pad location vs. heatmap peak
Power & runtime	Battery voltage vs. time @ 50 Hz

QUESTIONS FOR THE LAB

1. Paper type: instrumentation / validation / case series?
2. What tests are required to submit?
3. Reference standard: Tekscan? Other papers?
4. Do we need Human data? Or is bench testing enough?
5. Novelty: sensing · 3D map · low-cost · workflow?
6. Target venue for wearable/medical devices?
7. Next: lit gaps, IRB, co-authors, stats, figures?

THANK YOU.

LET'S REDEFINE THE FIT

Project90 · Prosthetic Smart Sock · University of Calgary

